



Midterm II Exam
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Electromagnetic Theory
Electrical Engineering

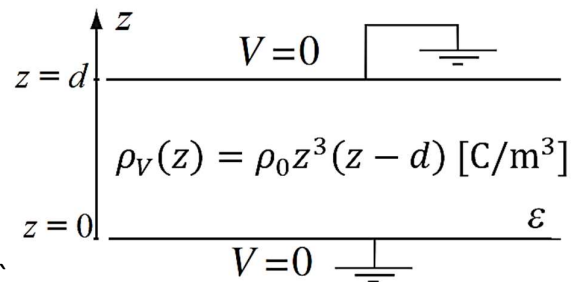
Spring 2021 **Points 100**
Habib University, KHI

INSTRUCTIONS:

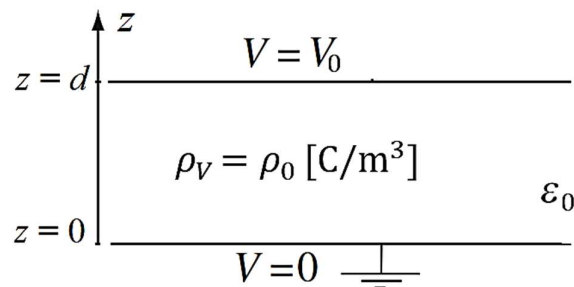
1. YOU ARE NOT ALLOWED TO DISCUSS WITH EACH OTHER.
2. DO NOT SHARE YOUR ROUGH/FAIR WORK WITH OR COPY/CHEAT FROM OTHERS; YOUR WORK MUST BE YOUR “INDEPENDENT” WORK.
3. NO PENCIL WORK, USE PROPER BLACK MARKER OR POINTER.
4. PROVIDE A NEAT WORK WHILE AVOIDING EXCESSIVE NUMBER OF PAGES.
5. ENCLOSE YOUR FINAL RESULTS IN RECTANGULAR BOXES.
6. SINGLE COLUMN ONLY IN PORTRAIT FORMAT. LEAVE SOME SPACES AS LEFT/RIGHT MARGINS.
7. IF REQUIRED, ADJUST CONTRAST LEVEL IN SCANS TO REMOVE DARK BACKGROUND.
8. IF REQUIRED, CROP YOUR IMAGES TO REMOVE UNNECESSARY SURROUNDINGS.

Question 1 [15]:

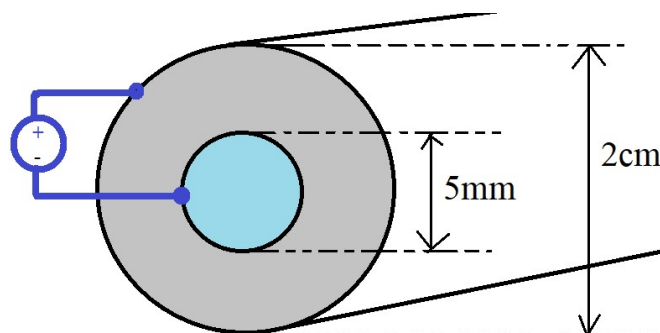
Consider a capacitor where (infinite) parallel plates are separated by a distance d [m]. The potential on each plate is zero (both are grounded). The dielectric between the plates has permittivity ϵ [F/m]. Within dielectric, the volume charge density is given by $\rho_V(z) = \rho_0 z^3(z - d)$ [C/m³]. Using the Poisson's equation, find the potential and electric field intensity in the region between the plates of the capacitor.

**Question 2 [15]:**

Consider a capacitor where (infinite) parallel plates are separated by a distance d [m]. The potential on the lower and upper plates are zero and V_0 , respectively. The dielectric between the plates has permittivity ϵ_0 [F/m]. Within dielectric, there is a volume charge density distributed according to $\rho_V = \rho_0$ [C/m³] (fixed). Using the Poisson's equation, find the potential and electric field intensity in the region between the plates of the capacitor.

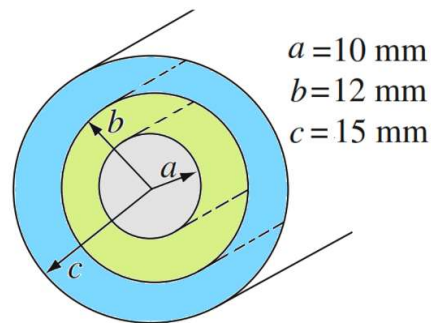
**Question 3 [20]:**

Consider a standard coaxial cable. The inner conductor is made of a solid wire, 5 mm in diameter, and the outer, thick conducting shell 2 cm in diameter is used. A direct voltage of 100V is connected with the positive pole connected to the outer conductor. Calculate: the potential distribution and electric field intensity everywhere in the coaxial cable.



Question 4 [10]:

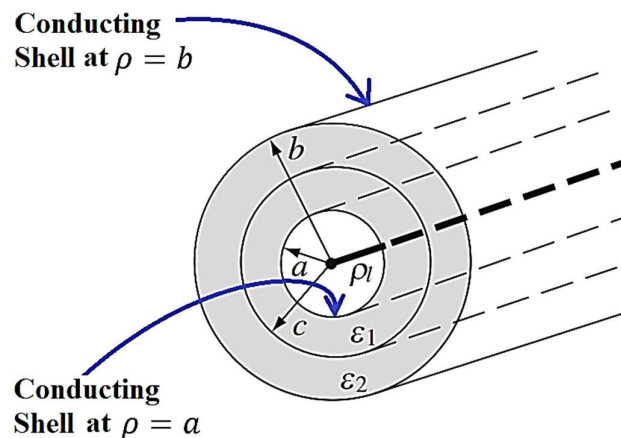
Consider a multi-layer conductor, 1 m long, as shown in Figure below.



The inner most layer is steel, the intermediate layer is aluminum, and the outer layer is copper, used for appearances' sake. The conductivities of aluminum, copper, and steel are 3.5×10^7 , 5.7×10^7 , and 1.0×10^7 Siemens/meter, respectively. Calculate the resistance of the conductor. You may use known formulas for resistance from the book.

Question 5 [40]:

Two long very thin conducting concentric cylindrical shells (at $\rho = a$ and $\rho = b$) are separated by two dielectric layers of permittivity ϵ_1 [F/m] and ϵ_2 [F/m] as shown in Figure below.



A line of charge with line charge density ρ_l [C/m] is placed at the center of the inner shell.

- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $0 < \rho < a$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $a < \rho < c$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $c < \rho < b$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $b < \rho < \infty$.

Note: Use Gauss's theorem in the evaluation of $\mathbf{E}$ or $\mathbf{D}$. Show all required steps in part (a). You may reuse some of the results already obtained in part (a) in other parts without showing all steps.

Q1

$$\nabla^2 V = \frac{-\rho_0}{\epsilon} = -\frac{(\rho_0 z^3(z-d))}{\epsilon}$$
$$= -\frac{\rho_0 z^4 + \rho_0 z^3 d}{\epsilon}$$

Using the $\nabla^2 V$ operator,

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} = \frac{-\rho_0 z^4}{\epsilon} + \frac{\rho_0 z^3 d}{\epsilon}$$

Integrating both sides,

$$\frac{\partial V}{\partial z} = \frac{-\rho_0 z^5}{5\epsilon} + \frac{\rho_0 d z^4}{4\epsilon} + C_1$$

Integrating both sides,

$$V = \frac{-\rho_0 z^6}{30\epsilon} + \frac{\rho_0 d z^5}{20\epsilon} + C_1 z + C_2 \quad \text{---(i)}$$

Applying the boundary conditions,

$$V(z=d)=0 \Rightarrow 0 = \frac{-\rho_0 d^6}{30\epsilon} + \frac{\rho_0 d^6}{20\epsilon} + C_1 d + C_2$$

$$0 = \frac{1}{60\epsilon} \rho_0 d^6 + C_1 d + C_2$$

$$\boxed{C_2 = -\frac{1}{60\epsilon} \rho_0 d^6 - C_1 d}$$

$$V(z=0)=0 \Rightarrow 0 = C_2$$

Then,

$$\boxed{C_1 = -\frac{1}{60\epsilon} \rho_0 d^5}$$

Substituting expressions for C_1 & C_2 into eq(i),

$$\boxed{V = \frac{-\rho_0 z^6}{30\epsilon} + \frac{\rho_0 d z^5}{20\epsilon} - \frac{1}{60\epsilon} \rho_0 d^5 z} \quad V$$

$$\vec{E} = -\nabla V$$

$$= - \left[\frac{\partial V}{\partial z} a_z \right] = - \left[\frac{-\rho_0 z^5}{5\epsilon} + \frac{\rho_0 d z^4}{4\epsilon} - \frac{\rho_0 d^5}{60\epsilon} \right] a_z$$

$$\vec{E} = \frac{\rho_0 z^5}{5\epsilon} - \frac{\rho_0 d z^4}{4\epsilon} + \frac{\rho_0 d^5}{60\epsilon} a_z$$

$$\boxed{\vec{E} = \frac{\rho_0}{\epsilon} \left[\frac{z^5}{5} - \frac{d z^4}{4} + \frac{d^5}{60} \right] a_z}$$

Q2

$$\nabla^2 V = \frac{-\rho_0}{\epsilon} = \frac{-\rho_0}{\epsilon_0}$$

Using the $\nabla^2 V$ operator,

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_0}{\epsilon_0} \quad (\text{since } V \text{ depends on } z)$$

Integrating both sides,

$$\frac{\partial V}{\partial z} = -\frac{\rho_0}{\epsilon_0} z + C_1$$

Integrating both sides,

$$V = -\frac{\rho_0}{2\epsilon_0} z^2 + C_1 z + C_2$$

Applying boundary conditions,

$$V(z=d) = V_0 \rightarrow V_0 = -\frac{\rho_0}{2\epsilon_0} d^2 + C_1 d + C_2$$

$$\boxed{C_2 = V_0 + \frac{\rho_0}{2\epsilon_0} d^2 - C_1 d}$$

$$V(z=0) = 0 \rightarrow$$

then,

$$\boxed{0 = C_2}$$

$$\boxed{C_1 = \frac{V_0 + \frac{\rho_0}{2\epsilon_0} d}{d}}$$

Then, V becomes,

$$\boxed{V = -\frac{\rho_0}{2\epsilon_0} z^2 + \left(\frac{V_0 + \frac{\rho_0}{2\epsilon_0} d}{d} \right) z}$$

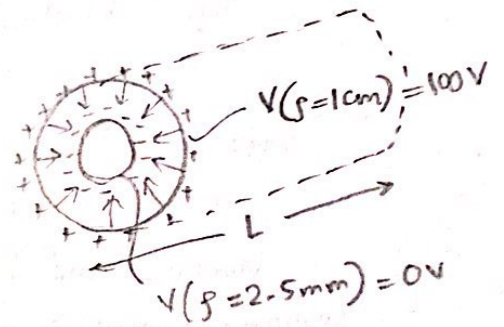
$$\begin{aligned}\bar{E} &= -\nabla V \\ &= -\left[\frac{\partial V}{\partial z} a_z\right] = -\left[\frac{-\rho_0 z}{\epsilon_0} + \frac{V_0}{d} + \frac{\rho_0 d}{2\epsilon_0}\right] a_z \\ &= \left[\frac{\rho_0 z}{\epsilon_0} - \frac{V_0}{d} - \frac{\rho_0 d}{2\epsilon_0}\right] a_z\end{aligned}$$

$$\boxed{\bar{E} = \left[\frac{\rho_0}{\epsilon_0}\left(z - \frac{d}{2}\right) - \frac{V_0}{d}\right] a_z}$$

Q3

Since V depends only on ρ , Laplace's equation in cylindrical coordinates becomes,

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0$$



Integrating both sides,

$$\rho \frac{\partial V}{\partial \rho} = c_1$$

$$\frac{\partial V}{\partial \rho} = \frac{c_1}{\rho}$$

Integrating both sides,

$$\int \frac{\partial V}{\partial \rho} = \int \frac{c_1}{\rho} \Rightarrow V = c_1 \ln(\rho) + c_2 \quad \text{--- (i)}$$

Applying the boundary conditions,

$$V(\rho = 1\text{cm}) = 100\text{V} \rightarrow 100 = c_1 \ln(1 \times 10^{-2}) + c_2$$

$$c_2 = 100 - c_1 \ln(1 \times 10^{-2})$$

$$V(\rho = 2.5 \times 10^{-3}) = 0\text{V} \rightarrow 0 = c_1 \ln(2.5 \times 10^{-3}) + 100 - c_1 \ln(1 \times 10^{-2})$$

$$0 = c_1 \ln\left(\frac{2.5 \times 10^{-3}}{1 \times 10^{-2}}\right) + 100$$

$$c_1 \ln\left(\frac{1}{4}\right) + 100 = 0$$

$$\boxed{c_1 = \frac{-100}{\ln(1/4)}}$$

then,

$$c_2 = 100 - \left(\frac{-100}{\ln(1/4)} \right) \ln(1 \times 10^{-2})$$

(4)

$$c_2 = 100 + 100 \frac{\ln(0.01)}{\ln(0.25)}$$

Substituting the expressions for c_1 & c_2 into eq (i),

$$V = \frac{-100}{\ln(0.25)} \ln(\rho) + 100 + 100 \frac{\ln(0.01)}{\ln(0.25)}$$

$$V = \frac{100}{\ln(0.25)} \left(\ln(0.01) - \ln(\rho) \right) + 100 \text{ V}$$

$$\vec{E} = -\nabla V$$

$$= - \left[\frac{\partial V}{\partial \rho} \right] \hat{a}_\rho = - \left[\frac{-100}{\ln(0.25)\rho} \right] \hat{a}_\rho$$

$$\vec{E} = \frac{100}{\ln(0.25)\rho} \hat{a}_\rho \text{ V/m}$$

Q4

Since V varies along the length of the conductor,

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} = 0$$

Integrating both sides,

$$\frac{\partial V}{\partial z} = c_1$$

Integrating both sides,

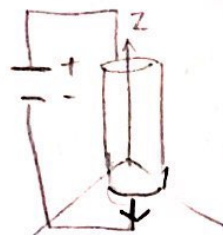
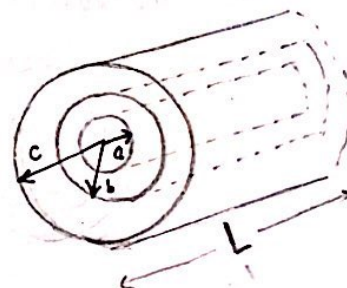
$$V = c_1 z + c_2$$

$$V(z=L) = V_0 \rightarrow V_0 = c_1 + c_2$$

$$c_1 = V_0 - c_2$$

$$V(z=0) = 0 \rightarrow 0 = c_2 \text{ Then, } c_1 = V_0$$

$$V = V_0 z$$



$$\vec{E} = -\nabla V$$

$$= -\frac{\partial V}{\partial z} \hat{a}_z = -V_0 \hat{a}_z$$

$$I = \int_S \sigma \vec{E} \cdot d\vec{s} \quad ; \quad d\vec{s} = -\rho d\rho d\phi \hat{a}_z$$

$$= \sigma \int (-V_0 \hat{a}_z) (-\rho d\rho d\phi \hat{a}_z)$$

$$= \sigma V_0 \int_{\phi=0}^{2\pi} \int_{\rho=0}^{10\text{mm}} \rho d\rho d\phi \quad (\text{for Steel})$$

$$= (1 \times 10^7) V_0 \int_{\phi=0}^{2\pi} \left. \frac{\rho^2}{2} \right|_{\rho=0}^{10\text{mm}} d\phi = (1 \times 10^7) V_0 (2\pi) \left(\frac{(10 \times 10^{-3})^2}{2} \right)$$

$$= \boxed{1000 \pi V_0}$$

For Aluminium,

$$I = \sigma V_0 \int_{\phi=0}^{2\pi} \int_{\rho=0}^{12\text{mm}} \rho d\rho d\phi = (3.5 \times 10^7) V_0 (2\pi) \left(\frac{(12 \times 10^{-3})^2}{2} \right)$$

$$= \boxed{5040 V_0 \pi}$$

For Copper,

$$I = \sigma V_0 \int_{\phi=0}^{2\pi} \int_{\rho=0}^{15\text{mm}} \rho d\rho d\phi = (5.7 \times 10^7) V_0 (2\pi) \left(\frac{(15 \times 10^{-3})^2}{2} \right)$$

$$= \boxed{12825 V_0 \pi}$$

$$R_{\text{steel}} = \frac{V_0}{I} = \frac{V_0}{1000 \pi V_0} = \frac{1}{1000 \pi}$$

$$R_{\text{aluminium}} = \frac{V_0}{I} = \frac{V_0}{5040 \pi V_0} = \frac{1}{5040 \pi}$$

$$R_{\text{copper}} = \frac{V_0}{I} = \frac{V_0}{12825 \pi V_0} = \frac{1}{12825 \pi}$$

$$R_{\text{total}} = R_{\text{steel}} \parallel R_{\text{aluminium}} \parallel R_{\text{copper}}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_{\text{steel}}} + \frac{1}{R_{\text{aluminium}}} + \frac{1}{R_{\text{copper}}}$$

⑥

$$\frac{1}{R_T} = \frac{1}{1000\Omega} + \frac{1}{5040\Omega} + \frac{1}{12825\Omega}$$

$$\frac{1}{R_T} = 18865\Omega$$

$\Rightarrow$

$$R_T = 16.9\mu\Omega$$

Question 5 [40]:

Two long very thin conducting concentric cylindrical shells (at $\rho = a$ and $\rho = b$) are separated by two dielectric layers of permittivity ϵ_1 [F/m] and ϵ_2 [F/m] as shown in Figure below.

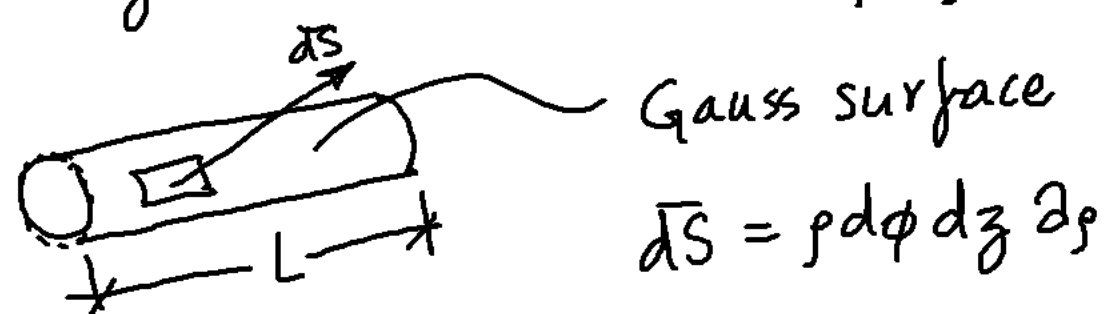
A line of charge with line charge density ρ_l [C/m] is placed at the center of the inner shell.

- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $0 < \rho < a$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $a < \rho < c$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $c < \rho < b$.
- Obtain the expressions for $\mathbf{E}$, $\mathbf{D}$ and $\mathbf{P}$ in the region $b < \rho < \infty$.

Solution

- For $0 < \rho < a$, the space is empty but contains a line charge density ρ_l .

We may comfortably assume that $\mathbf{E} = E_\rho \hat{\rho}$ or $\mathbf{D} = D_\rho \hat{\rho}$



$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = \rho_l L$$

$$D_\rho \rho \int d\phi dz = 2\pi L \rho D_\rho = \rho_l L \Rightarrow D_\rho = \frac{\rho_l}{2\pi \rho}$$

$$\mathbf{D} = D_\rho \hat{\rho}$$

$$\mathbf{E} = \frac{D_\rho}{\epsilon_0} \hat{\rho} = \frac{\rho_l}{2\pi \epsilon_0 \rho} \hat{\rho}$$

Since this region is free space
 $\mathbf{P} = 0$.

(c) $c < \rho < b$

Again $\mathbf{D}$ remains the same as there is no free charge at the boundary $\rho = c$. Also there is no tangential component of $\mathbf{D}$.

$$\mathbf{D} = \frac{\rho_l \hat{\rho}}{2\pi \rho} \Rightarrow \mathbf{E} = \frac{\mathbf{D}}{\epsilon_2} = \frac{\rho_l \hat{\rho}}{2\pi \epsilon_2 \rho}$$

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} = \frac{\rho_l \hat{\rho}}{2\pi \rho} \left(1 - \frac{\epsilon_0}{\epsilon_2}\right)$$

The units of $\mathbf{D}$ and $\mathbf{P}$ are C/m².

The unit of $\mathbf{E}$ is V/m.

(b) $a < \rho < c$

This is a material space with permittivity ϵ ,

$$\text{So } \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

$$\mathbf{P} = \epsilon_0 \chi_{e1} \mathbf{E}$$

$$\mathbf{D} = \epsilon_0 (1 + \chi_{e1}) \mathbf{E} = \epsilon_0 \epsilon_{r1} \mathbf{E} = \epsilon_1 \mathbf{E}$$

Note that the expression of $\mathbf{D}$ as obtained in part (a) is applicable here without any change for three reasons.

(A) the result obtained in (a) does not depend on any material space.

(B) There is no free charge at the boundary $\rho = a$.

(C) There is no tangential component in $\mathbf{D}$

$$\text{So } \mathbf{D} = \frac{\rho_l \hat{\rho}}{2\pi \rho}$$

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_1} = \frac{\rho_l \hat{\rho}}{2\pi \epsilon_1 \rho}$$

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} = \frac{\rho_l \hat{\rho}}{2\pi} \left(1 - \frac{\epsilon_0}{\epsilon_1}\right)$$

(d) $\rho > b$

There is no free charge at $\rho = b$.

$\mathbf{D}$ remains the same.

The region is empty space

$$\text{So } \mathbf{P} = 0 \text{ and } \mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \frac{\rho_l \hat{\rho}}{2\pi \epsilon_0 \rho}$$